LAB ASSIGNMENT NO. 1

Aim:

To execute and understand the working of basic **Data Definition Language (DDL)** commands — **CREATE**, **ALTER**, **DROP**, and **TRUNCATE** — using SQL on given tables.

Objectives:

1. To create and define the structure of database tables using the CREATE TABLE command.
2. To modify the structure of existing tables using the ALTER TABLE command.
3. To delete the complete structure of tables using the DROP TABLE command.
4. To remove all data from tables without affecting the structure using the TRUNCATE TABLE command.
5. To understand constraints such as **PRIMARY KEY**, **CHECK**, **DEFAULT**, and **FOREIGN KEY**.

Theory:

DDL (Data Definition Language) commands are used to define, modify, and remove database structures but not the data itself.

The key DDL commands are:

- **CREATE TABLE:** Used to create a new table in the database.
- **ALTER TABLE:** Used to add, modify, or delete columns or constraints in an existing table.
- **DROP TABLE:** Removes the entire table along with its structure and data permanently.
- **TRUNCATE TABLE:** Removes all records from a table but keeps the structure for future use.

Syntax:

CREATE TABLE table_name (column_name datatype [constraints]);
ALTER TABLE table_name ADD/MODIFY/DROP column_name datatype;
DROP TABLE table_name;
TRUNCATE TABLE table_name;

SQL Queries:

1. Creating the Item Table

```
CREATE TABLE Item (
   ItemCode CHAR(2) PRIMARY KEY,
   ItemName VARCHAR(20) NOT NULL,
   UnitMeasure VARCHAR(10) CHECK (UnitMeasure IN ('Mtr', 'Kg', 'Tonn', 'Ltr', 'Nos')),
   Rate FLOAT CHECK (Rate > 0),
   Stock FLOAT,
   Max_Stock FLOAT,
   Min_Stock FLOAT,
   DrawingsSp VARCHAR(10) DEFAULT 'DR1000',
   CHECK (Stock BETWEEN Min_Stock AND Max_Stock)
);
```

2. Creating the Trans Table

```
CREATE TABLE Trans (
   TType CHAR(2) CHECK (TType IN ('R1', 'R2', 'I1', 'I2')),
   TNO VARCHAR(4) PRIMARY KEY,
   TDate DATE,
   TItemCode CHAR(2) REFERENCES Item(ItemCode),
   TRate FLOAT,
   TQty FLOAT,
   TDisc FLOAT
);
```

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3. Using ALTER TABLE command

(a) Add a new column to the Item table:

```
ALTER TABLE Item ADD SupplierName VARCHAR(25);
```

(b) Modify data type of the column:

```
ALTER TABLE Item MODIFY Rate DECIMAL(10,2);
```

(c) Rename a column (syntax may vary by SQL dialect):

```
ALTER TABLE Item RENAME COLUMN DrawingsSp TO DrawingSpec;
```

(d) Drop a column from the table:

ALTER TABLE Trans DROP COLUMN TDisc;

---

4. Using TRUNCATE command

To delete all rows from Trans table without removing its structure:

TRUNCATE TABLE Trans;

---

5. Using DROP command

To permanently delete the Trans table:

DROP TABLE Trans;

Sample Output (Text Description):

- On executing the CREATE TABLE statements, the message **"Table created successfully."** appears.
- After applying ALTER TABLE, the message **"Table altered."** confirms modification.
- TRUNCATE TABLE removes all records but keeps the table.
- DROP TABLE deletes the table entirely — any further reference gives an **"Invalid object name"** error.

Result:

The given DDL commands (CREATE, ALTER, TRUNCATE, and DROP) were successfully executed on the tables **Item** and **Trans**, demonstrating how SQL defines and manages the structure of database objects.

Conclusion:

By executing the above SQL commands, we learned how to create and modify database structures using DDL commands. The experiment demonstrated the importance of constraints, data integrity, and referential relationships between tables.

LAB ASSIGNMENT NO. 2

Aim:

To execute and understand the working of basic **Data Manipulation Language (DML)** commands — **INSERT**, **SELECT**, **UPDATE**, and **DELETE** — using SQL on the given database tables.

Objectives:

1. To insert records into database tables using the INSERT INTO command.
2. To retrieve and display data using the SELECT command.
3. To modify existing data using the UPDATE command.
4. To remove records from tables using the DELETE command.
5. To understand how DML operations interact with constraints and relationships between tables.

Theory:

Data Manipulation Language (DML) commands are used to manipulate the data stored in database tables.
They operate on the data but do not affect the structure of the tables.

Common DML Commands:

Command	Description
INSERT	Adds new rows of data into a table.
SELECT	Retrieves specific data from one or more tables.
UPDATE	Modifies existing data in one or more rows.
DELETE	Removes specific rows from a table.

Syntax:

```
INSERT INTO table_name (column1, column2, ...) VALUES (value1, value2, ...);
SELECT column1, column2 FROM table_name WHERE condition;
UPDATE table_name SET column1 = value1 WHERE condition;
DELETE FROM table_name WHERE condition;
```

SQL Queries:

1. Inserting Records into Item Table

INSERT INTO Item (ItemCode, ItemName, UnitMeasure, Rate, Stock, Max_Stock, Min_Stock)
VALUES ('I1', 'Copper Wire', 'Mtr', 120.50, 200, 500, 50);

INSERT INTO Item (ItemCode, ItemName, UnitMeasure, Rate, Stock, Max_Stock, Min_Stock)
VALUES ('I2', 'Steel Rod', 'Kg', 95.00, 100, 300, 25);

2. Inserting Records into Trans Table

INSERT INTO Trans (TType, TNO, TDate, TItemCode, TRate, TQty, TDisc)
VALUES ('R1', 'T001', '2025-10-05', 'I1', 120.50, 50, 5);

INSERT INTO Trans (TType, TNO, TDate, TItemCode, TRate, TQty, TDisc)
VALUES ('I1', 'T002', '2025-10-06', 'I2', 95.00, 20, 2);

---

3. Retrieving Data Using SELECT Command

(a) Display all records from Item table:

SELECT * FROM Item;

(b) Display ItemCode, ItemName, and Rate only:

SELECT ItemCode, ItemName, Rate FROM Item;

(c) Display all transactions of type 'R1':

SELECT * FROM Trans WHERE TType = 'R1';

(d) Display ItemName and Stock where Stock < 150:

SELECT ItemName, Stock FROM Item WHERE Stock < 150;

(e) Display combined details from both tables using a simple join:

```
SELECT
    T.TNO,T.TDate, I.ItemName, T.TQty, T.TRate, T.TDisc
FROM
    Item I, Trans T
WHERE
    I.ItemCode = T.TItemCode;
```

Explanation:
This query retrieves data from both tables — `Item` and `Trans` — by matching the `ItemCode` in `Item` with `TItemCode` in `Trans`. It displays meaningful, related data such as transaction number, date, item name, quantity, rate, and discount in a single combined view.

4. Updating Data Using UPDATE Command

(a) Update rate of item 'I2' to 105:

```
UPDATE Item SET Rate = 105 WHERE ItemCode = 'I2';
```

(b) Increase stock by 50 for item 'I1':

```
UPDATE Item SET Stock = Stock + 50 WHERE ItemCode = 'I1';
```

(c) Update discount of transaction 'T002' to 3%:

```
UPDATE Trans SET TDisc = 3 WHERE TNO = 'T002';
```

5. Deleting Records Using DELETE Command

(a) Delete a record from Trans table:

```
DELETE FROM Trans WHERE TNO = 'T001';
```

(b) Delete items whose stock is below minimum level:

```
DELETE FROM Item WHERE Stock < Min_Stock;
```

Sample Output (Text Description):

- On executing INSERT commands, messages like **"1 row inserted."** are displayed.
- SELECT statements show tabular data of items and transactions.
- UPDATE confirms with **"1 row updated."**
- DELETE confirms with **"1 row deleted."**
- Integrity constraints prevent invalid entries such as negative rate or incorrect unit measure.

Result:
The DML commands **INSERT**, **SELECT**, **UPDATE**, and **DELETE** were successfully executed on the **Item** and **Trans** tables. Data manipulation operations were verified through retrieval queries.

Conclusion:
This experiment demonstrated how DML commands are used to manage data in relational database tables. Students learned how to add, view, modify, and delete data while maintaining integrity constraints and relational consistency.